

In [3]: `from sklearn.feature_extraction.text import TfidfVectorizer`

```
# Sample documents
documents = [
    "this is a sample document",
    "this document is another example",
    "tf idf is important in text processing"
]

# Create TF-IDF Vectorizer
vectorizer = TfidfVectorizer()

# Transform documents into TF-IDF matrix
tfidf_matrix = vectorizer.fit_transform(documents)

# Get feature names
feature_names = vectorizer.get_feature_names_out()

# Convert to array for display
print("Feature Names:\n", feature_names)
print("\nTF-IDF Matrix:\n", tfidf_matrix.toarray())
```

Feature Names:

```
['another' 'document' 'example' 'idf' 'important' 'in' 'is' 'processing'
 'sample' 'text' 'tf' 'this']
```

TF-IDF Matrix:

```
[[0.          0.4804584  0.          0.          0.          0.
  0.37311881  0.          0.63174505  0.          0.          0.4804584 ]
 [0.53409337 0.40619178 0.53409337 0.          0.          0.
  0.31544415 0.          0.          0.          0.          0.40619178]
 [0.          0.          0.          0.39687454 0.39687454 0.39687454
  0.2344005  0.39687454 0.          0.39687454 0.39687454 0.          ]]
```

In []: